EEE 413 INTRODUCTION TO VLSI FINAL PROJECT ASSIGNMENT

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Introduction:

In this project we've succesfully designed two different adders (Manchester carry chain & Carry Skip Adders) and compared them. The first adder which is MCCA is an adder controlled by a logic element called PG block. In this particular adder PG block is responsible of controlling the MOSFETS that either allows or prevents the propagation of the carry to the output. In addition, an XOR gate is responsible of the addition of two bits. MCCA adders are made out of MCC slices. Each slice can add two bits and create one carry bit according to the state of the PG block.

The working principle of the carry adder is that carry can be skip to the following block. It let us to decrease gate delay and create a path with no delay in propagation. It is also created to increase the speed of the ripple carry adder chain. 16B CSKA (Carry Skip Adder) was created by combining 2B CSKA blocks in this experiment.

The difference between the two adders is that MCCA is synchonous and CSKA is not.

SCHEMATICS & LAYOUTS

Carry Skip Adder:

OR:

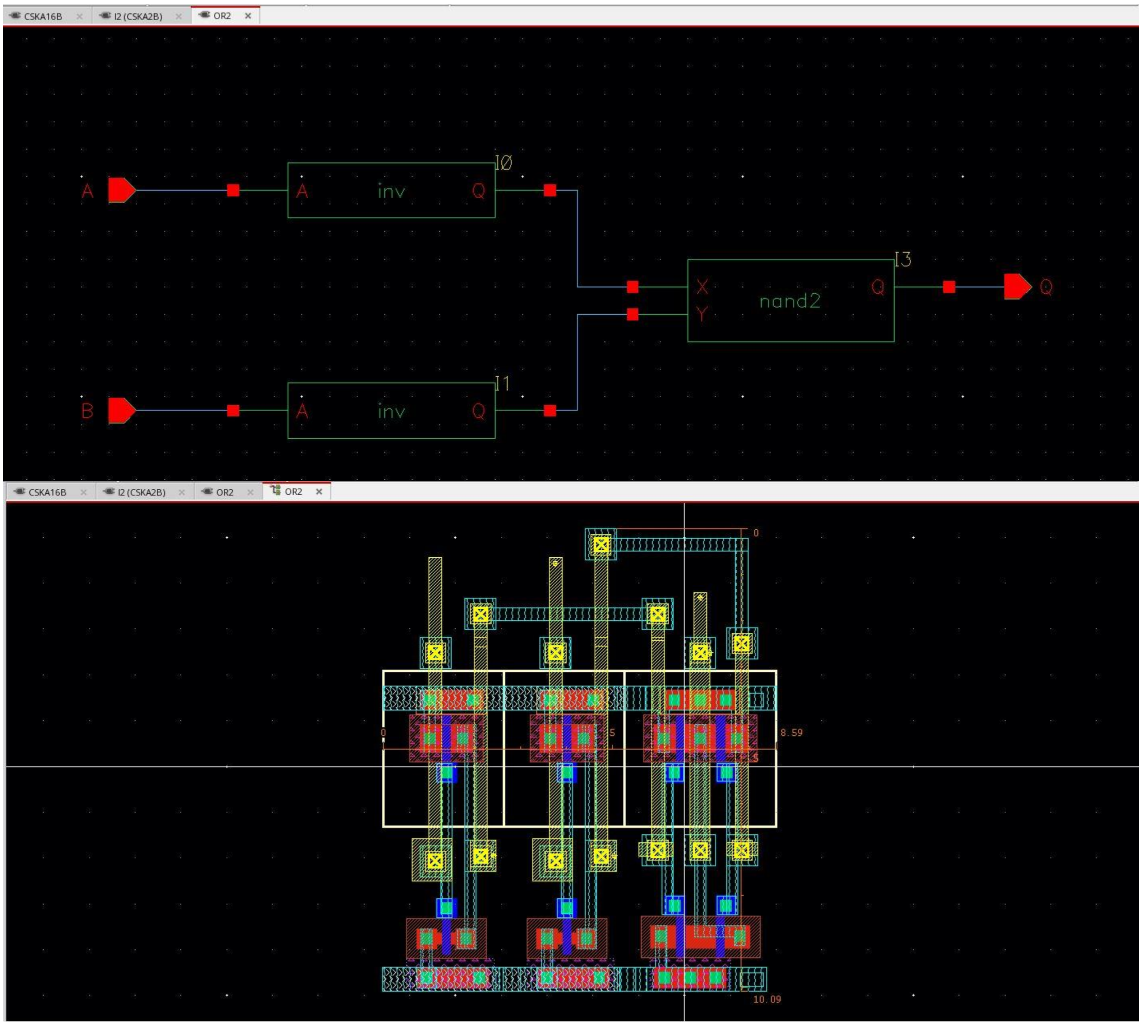


Figure 1 - OR Gate schematic & layout

XOR:

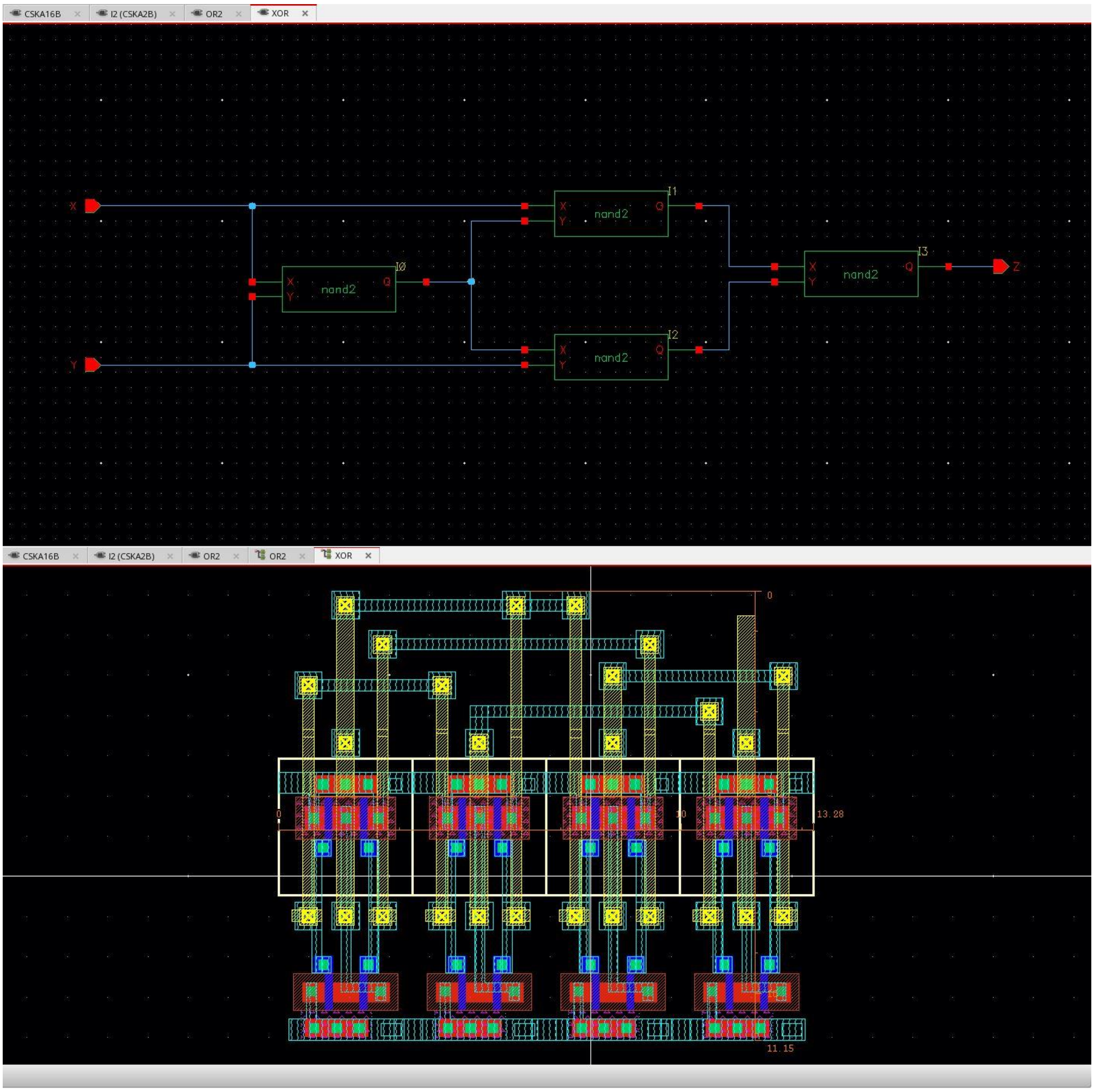


Figure 2 - XOR Gate schematic & layout

MUX:

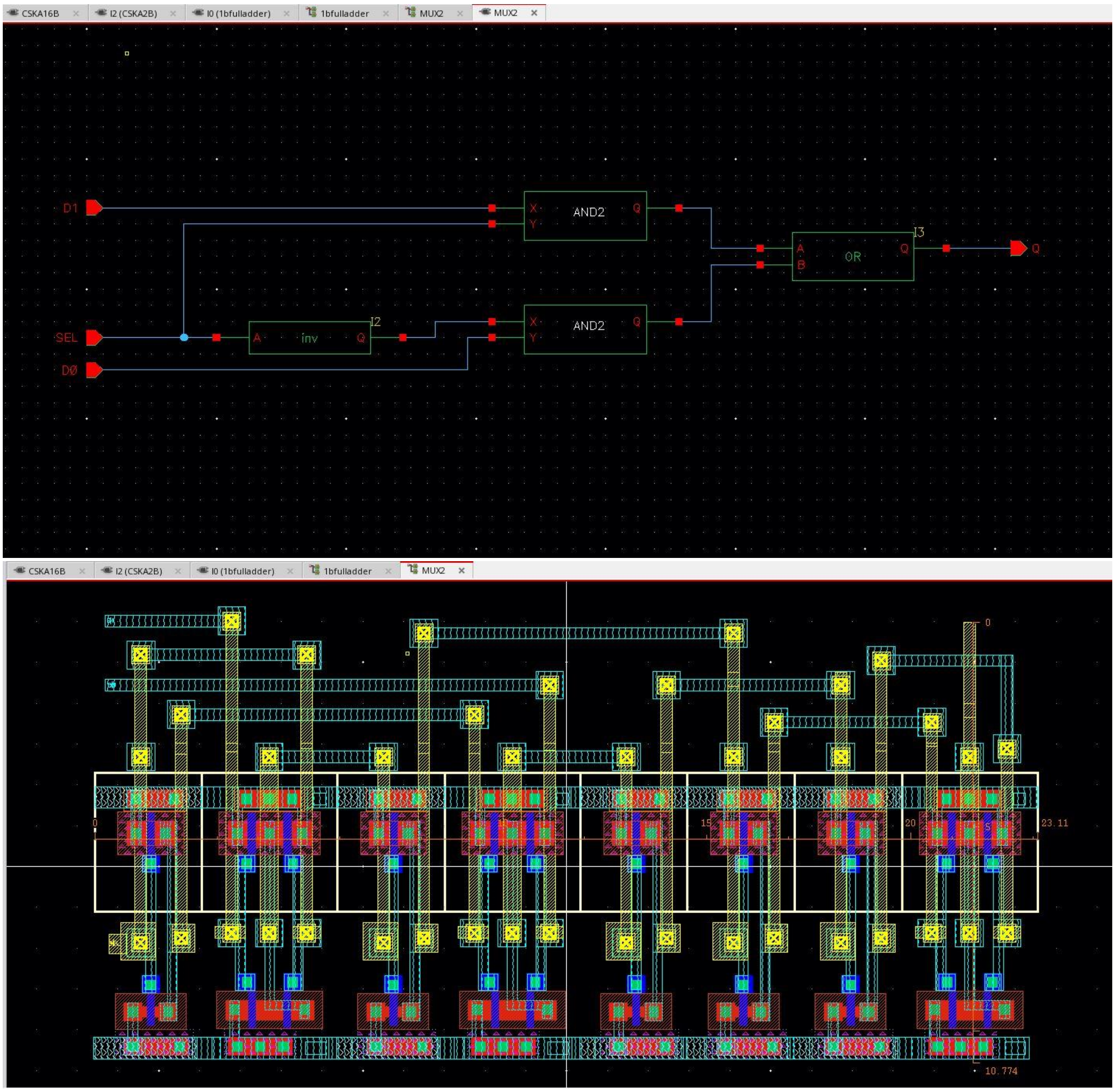


Figure 3 - MUX schematic & layout

1 BIT FULL ADDER:

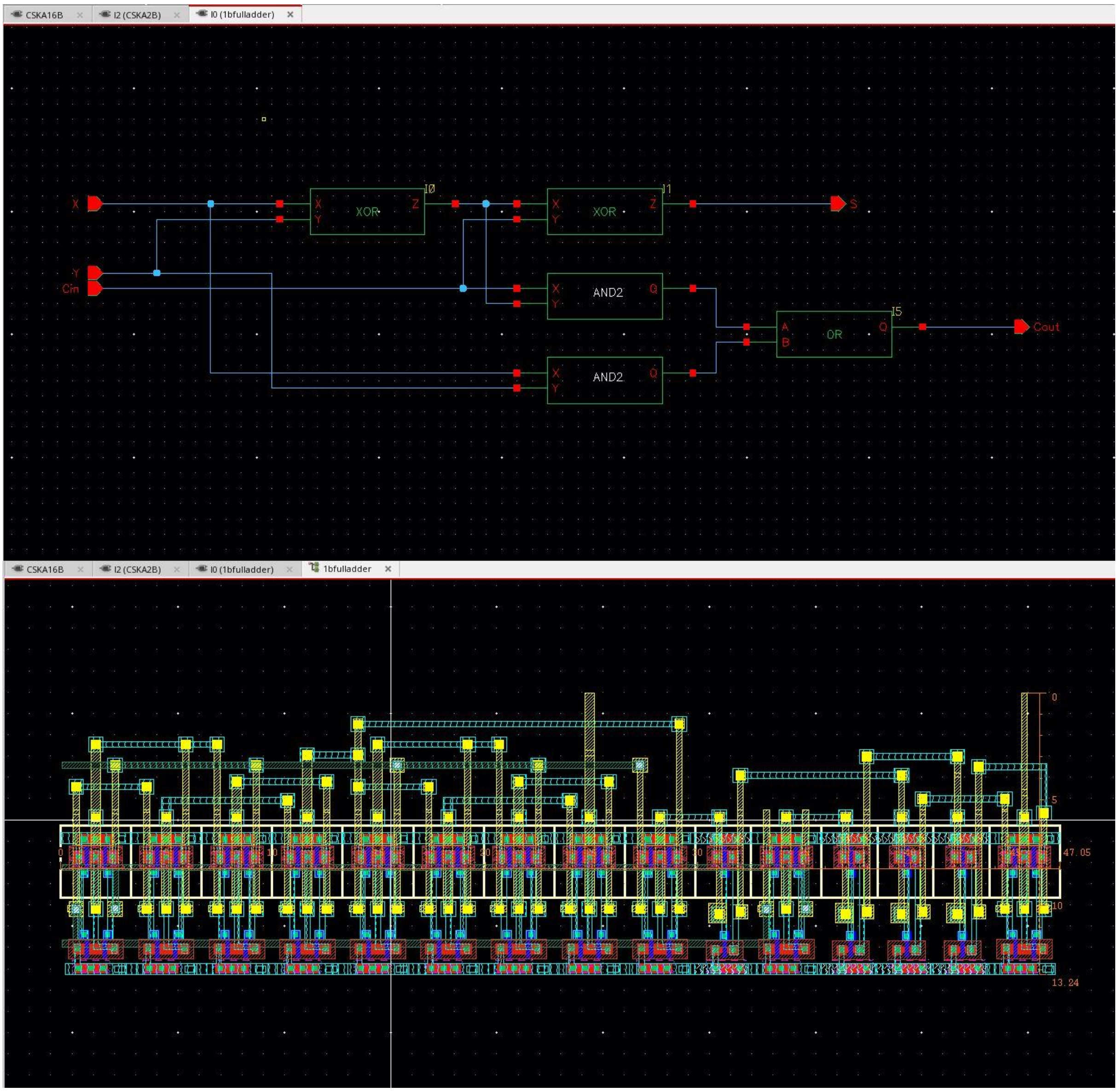


Figure 4 - 1 bit full adder schematic & layout

2 BIT FULL ADDER:

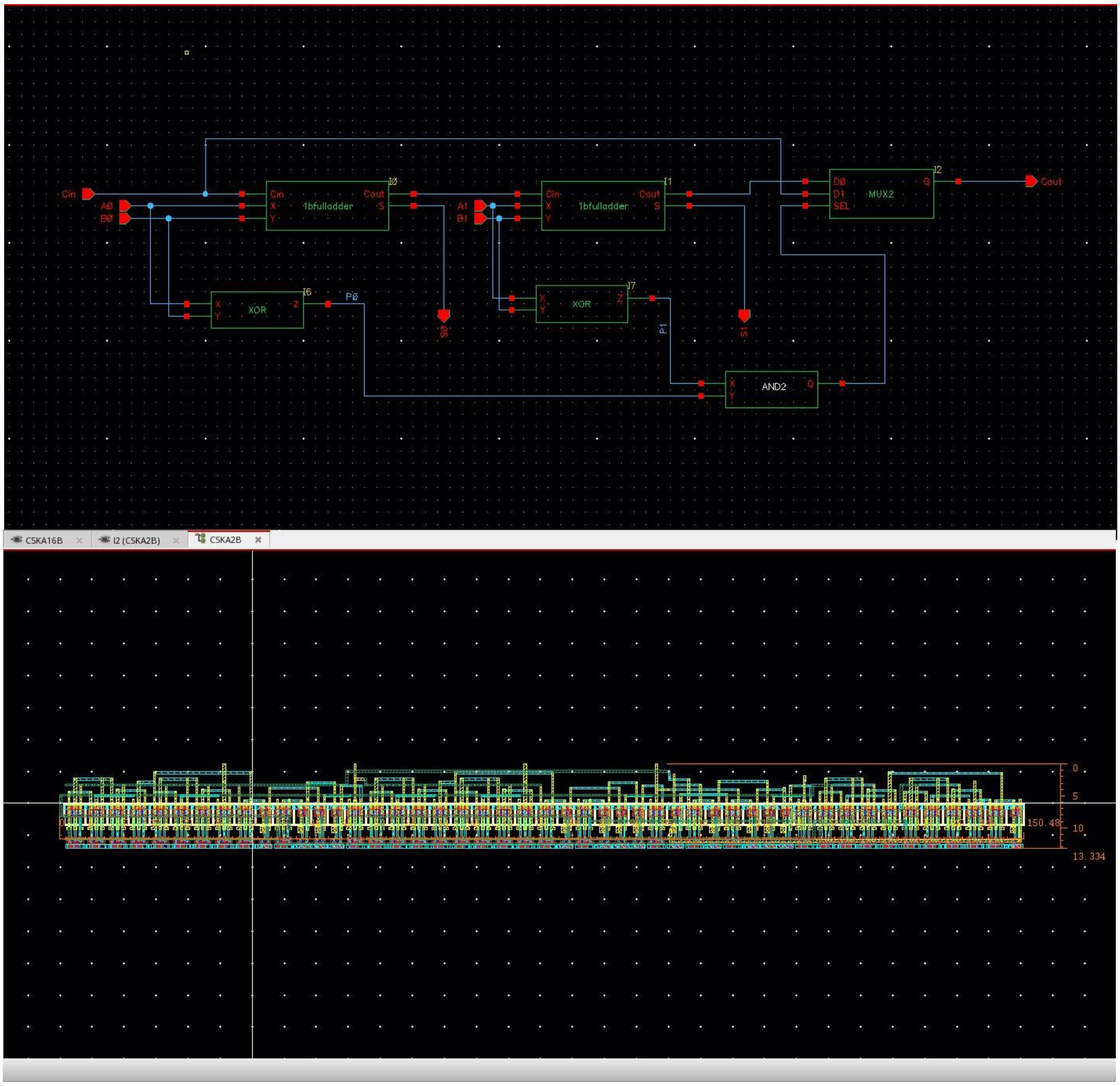


Figure 5 - 2 Bit Full Adder schematic & layout

16 BIT FULL ADDER:

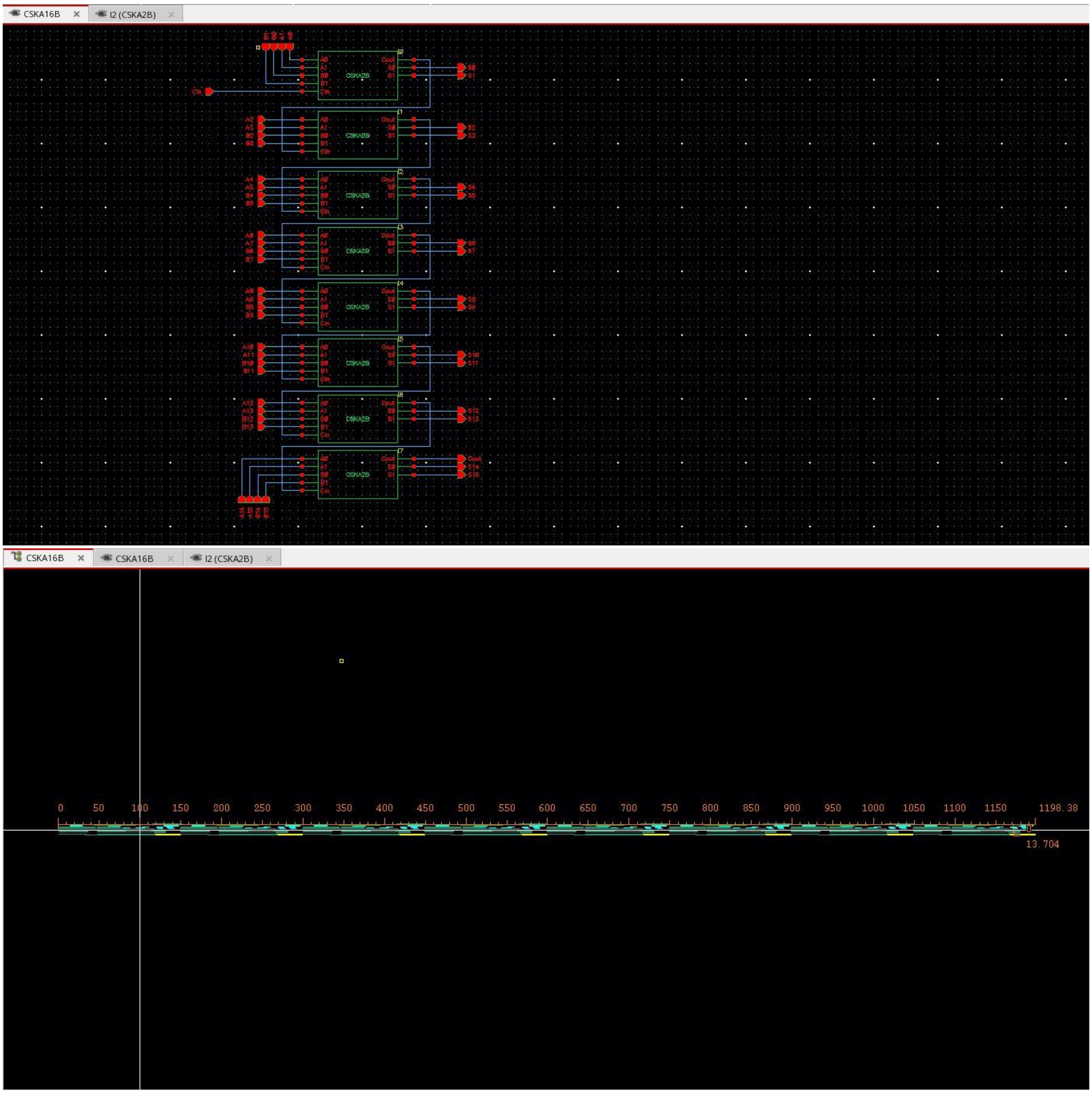


Figure 6 - 16 Bit Full Adder schematic & layout

Manchester Carry Chain Adder:

Buffer:

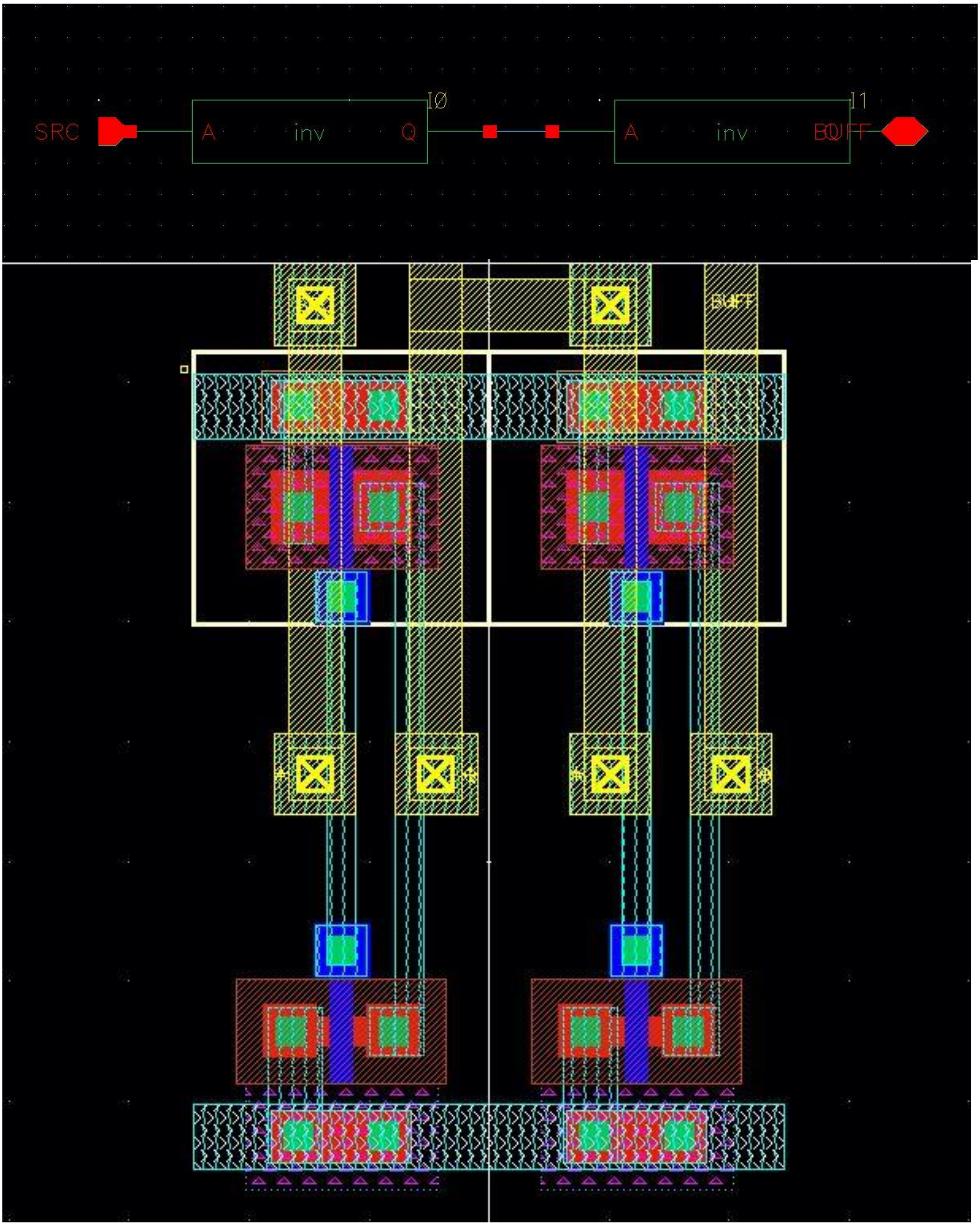


Figure 7 - Buffer schematic & layout

XOR:

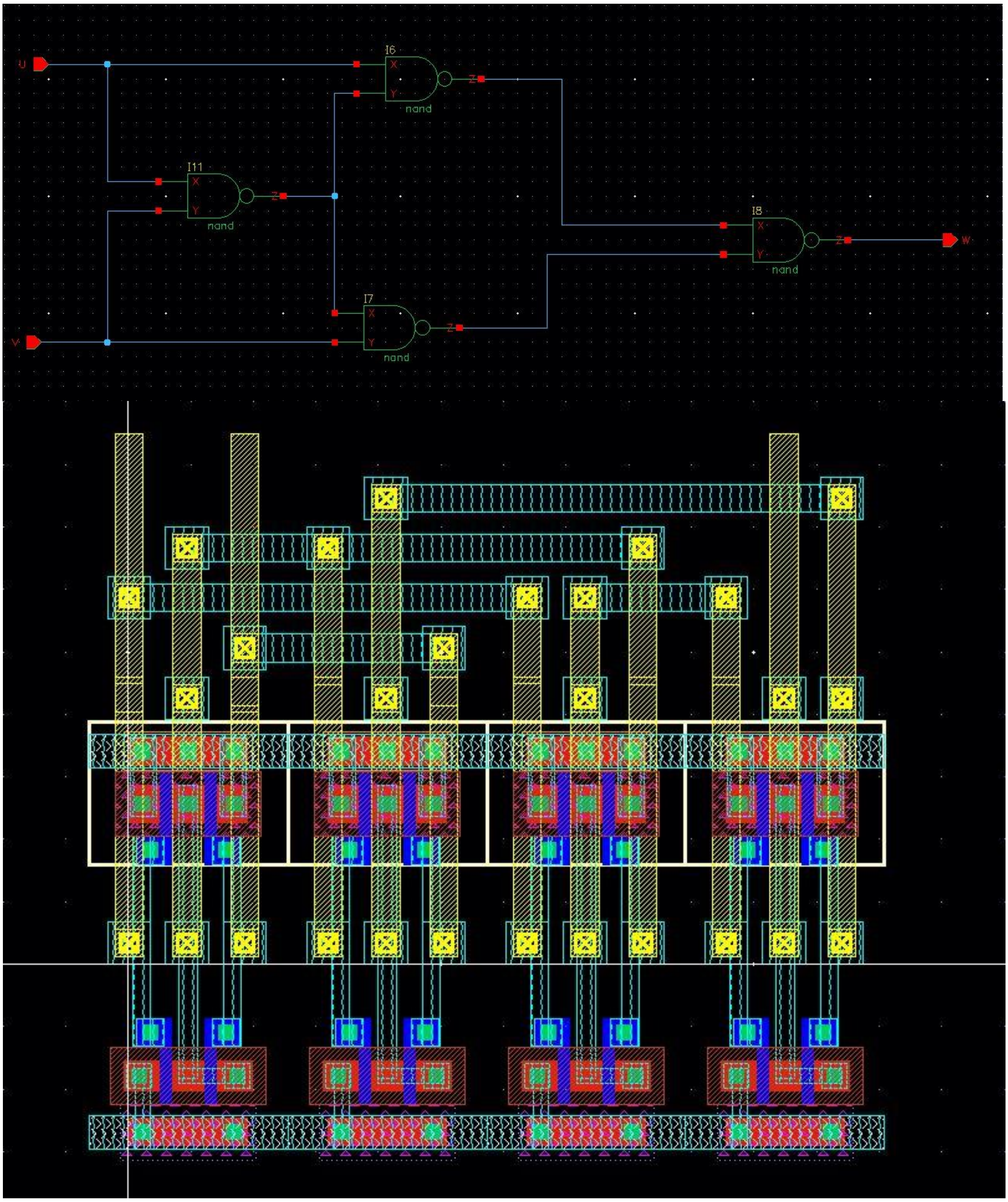


Figure 8 - XOR schematic & layout

PG Block:

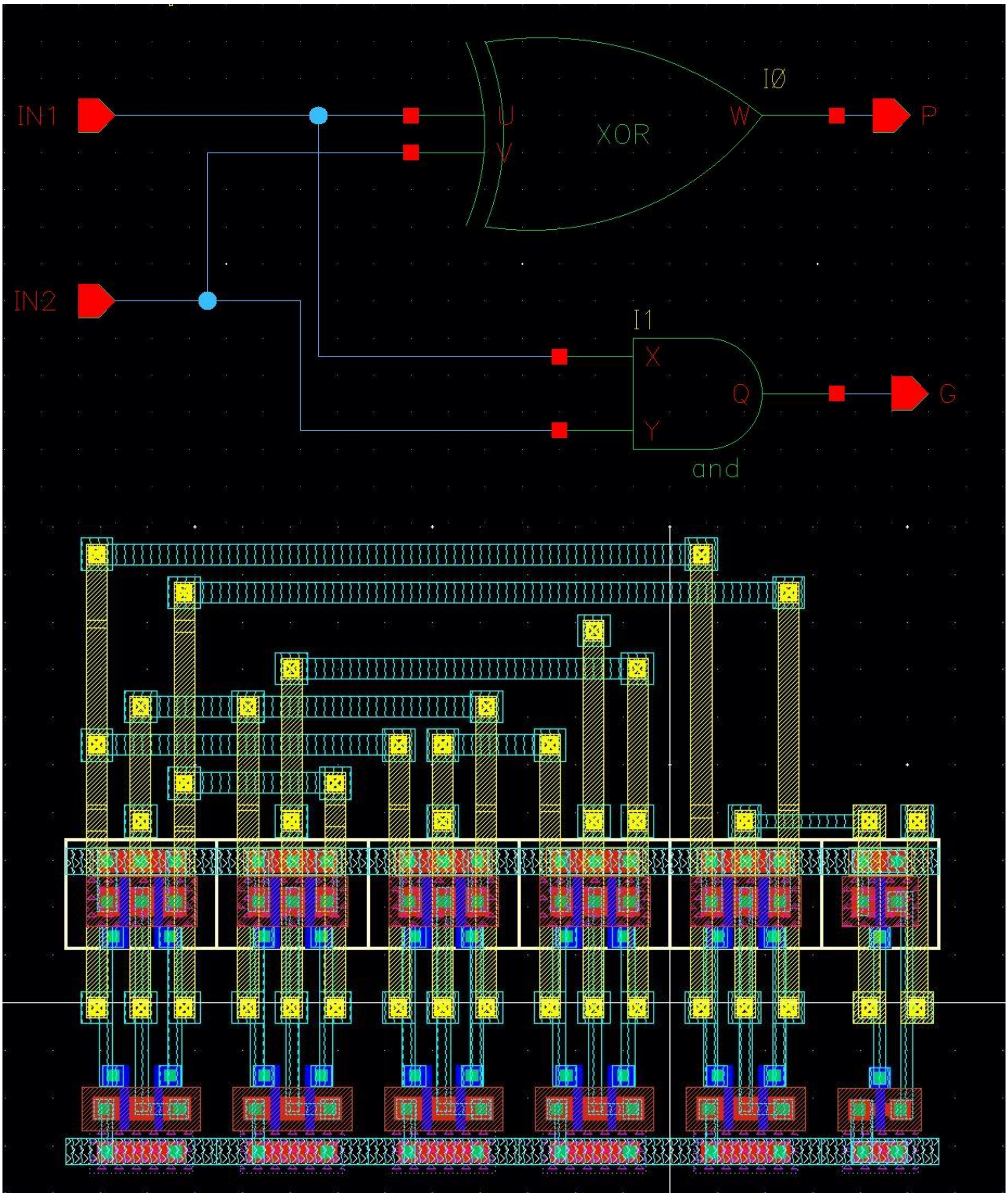


Figure 9 - PG Block schematic & layout

MCCA Slice:

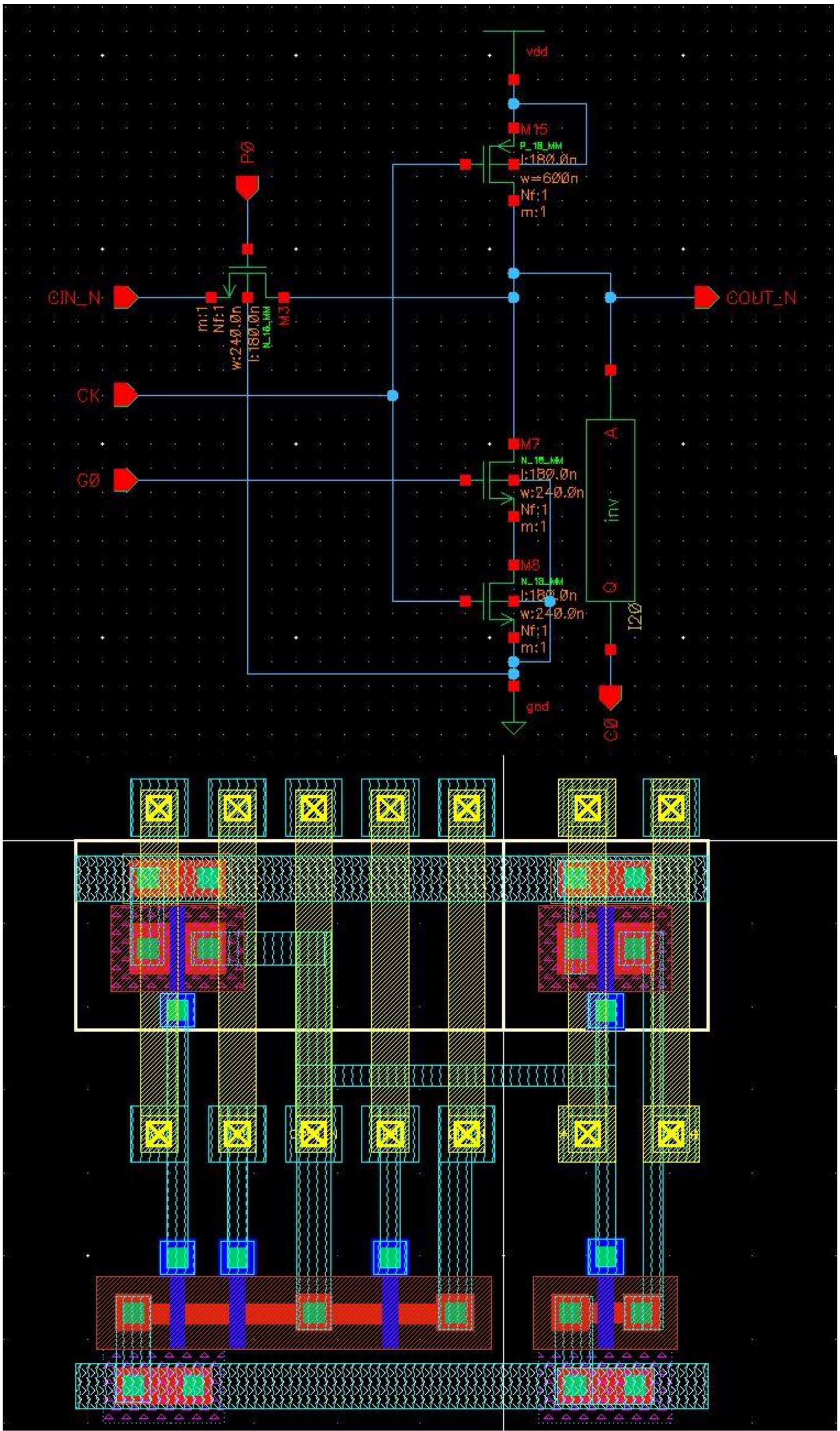


Figure 10 - MCCA schematic & layout

4 Bit MCCA

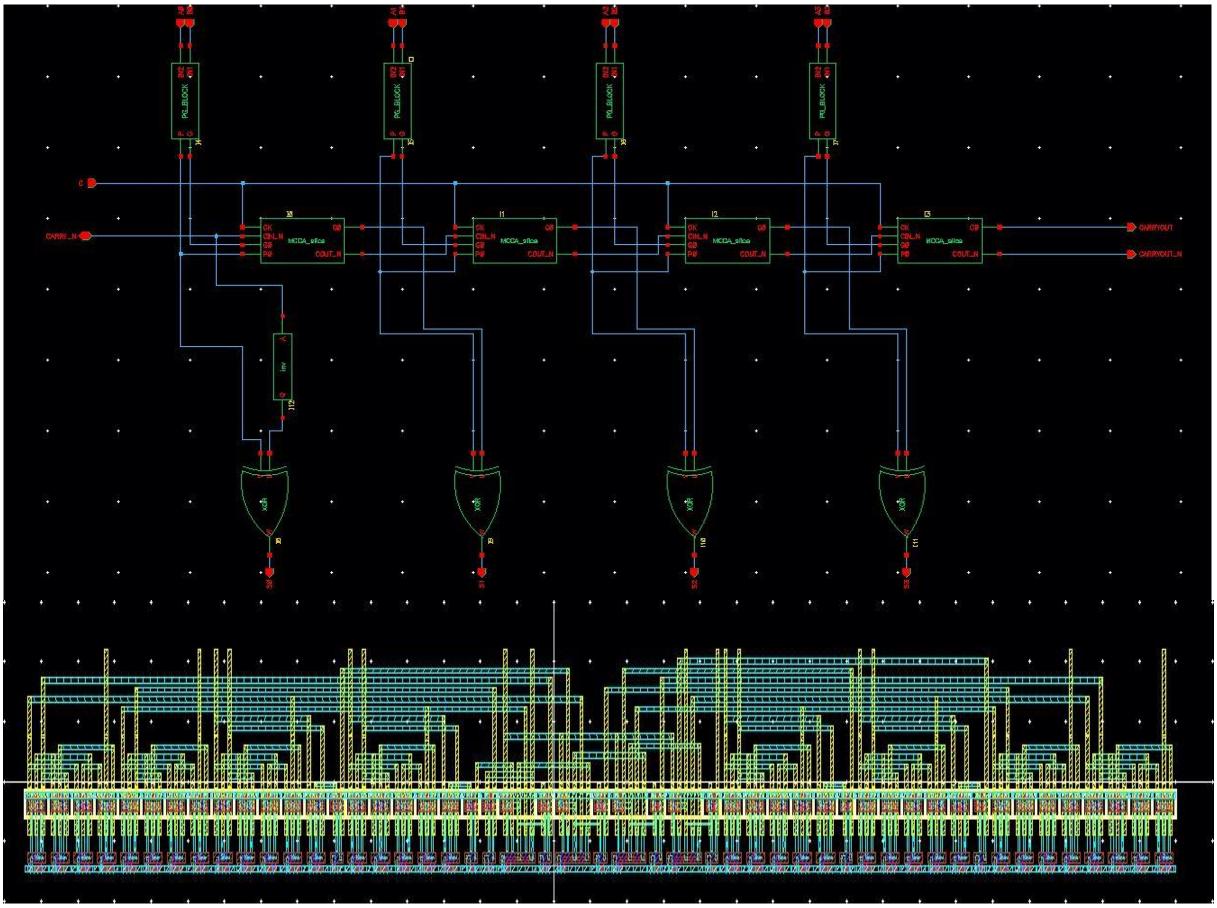


Figure 11 - 4 bit mcca schematic & layout

16 Bit MCCA:

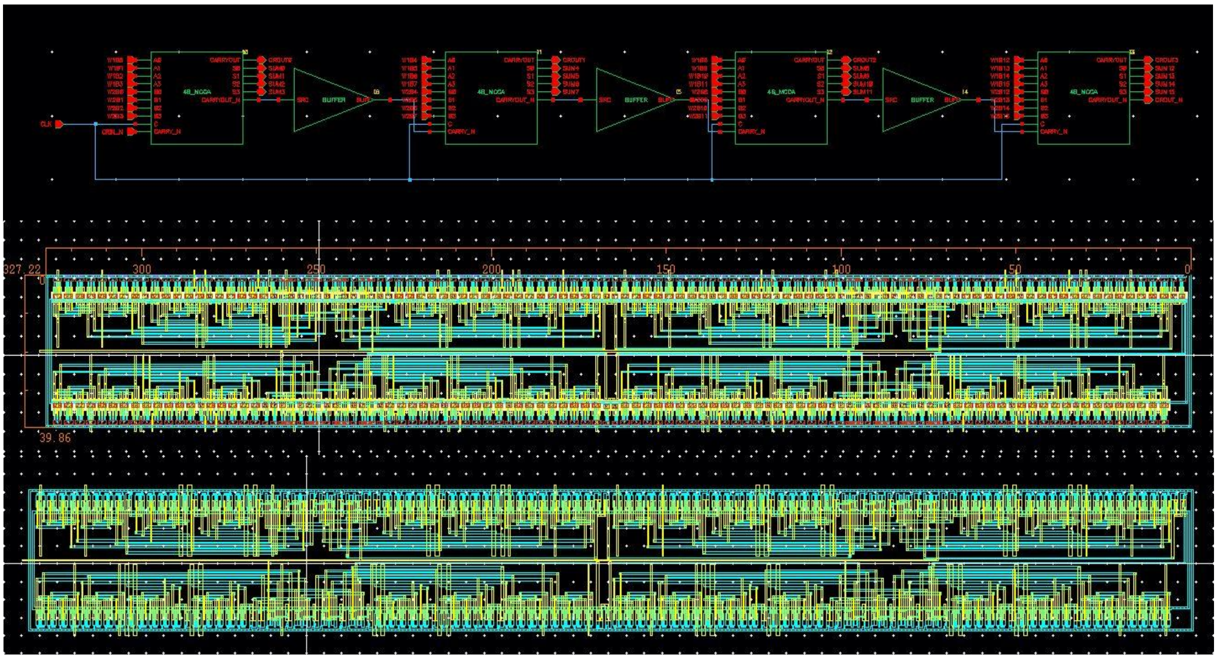
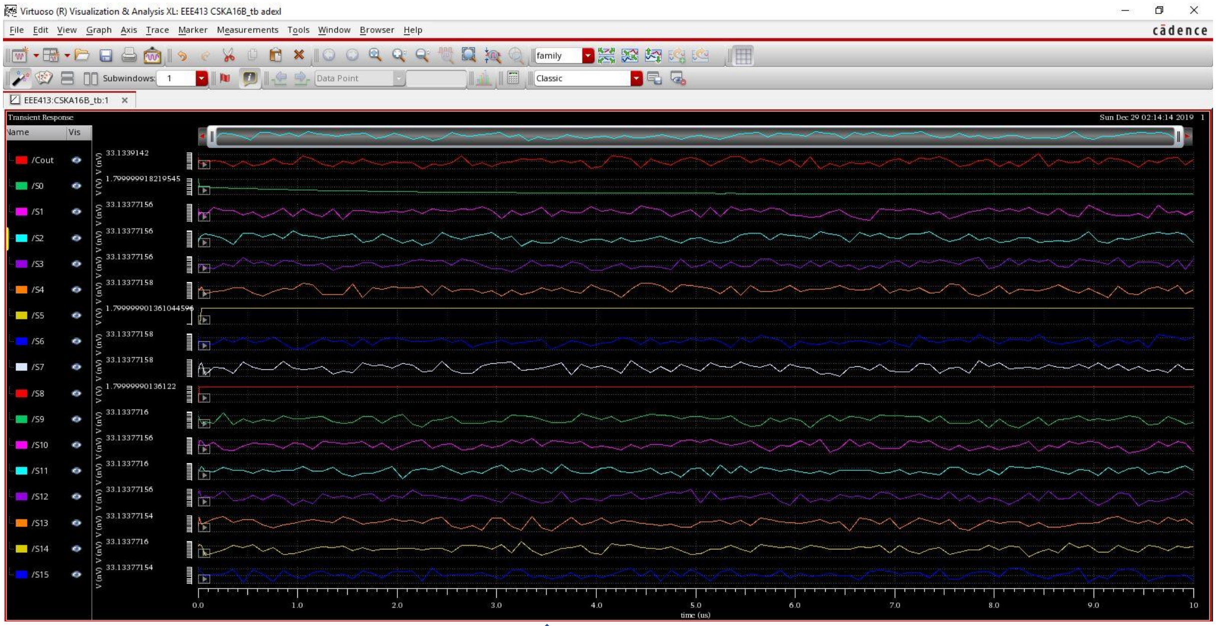
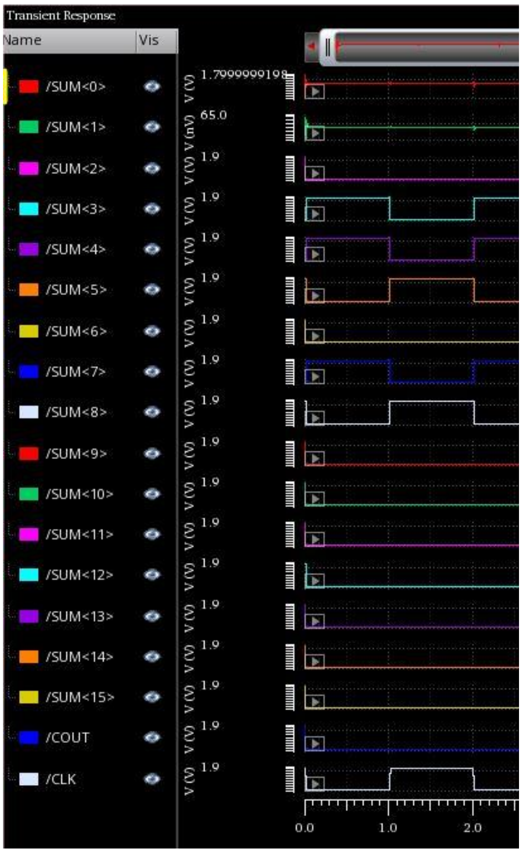


Figure 12 - 16 bit mcca schematic, layout & av_extracted view

Functionality:



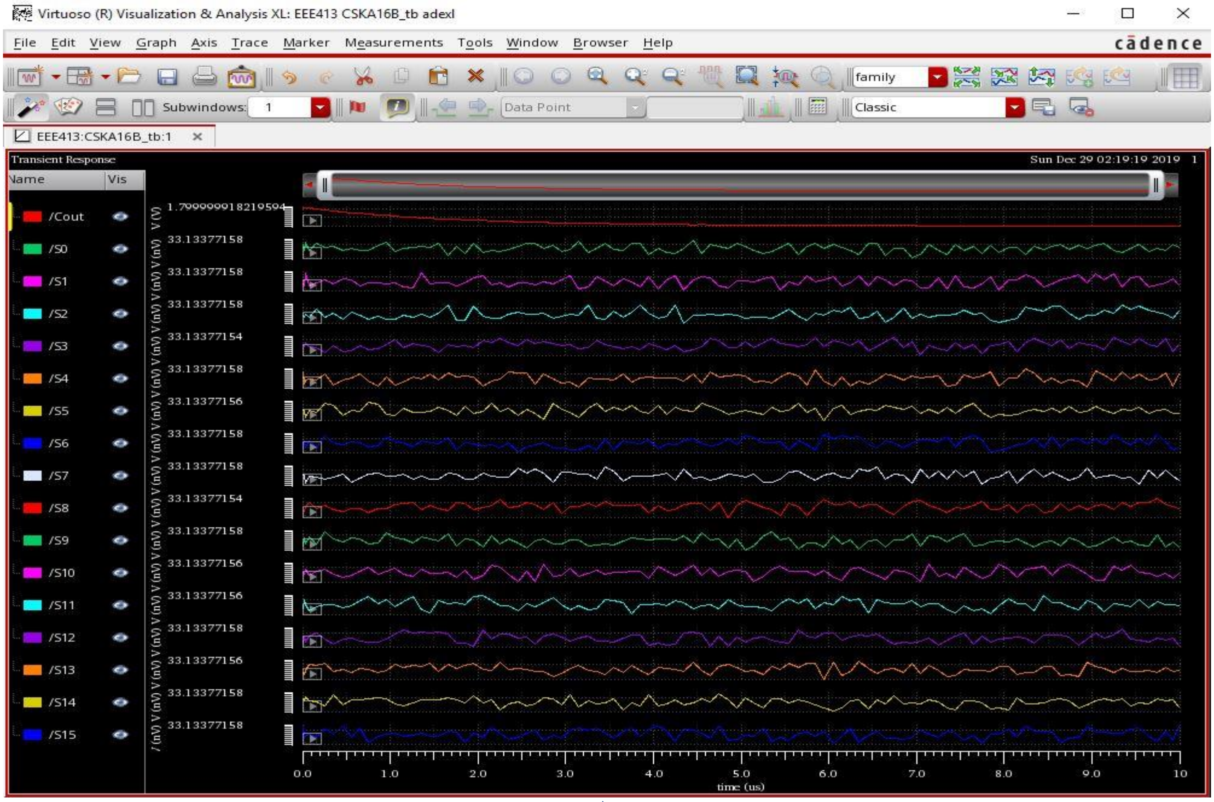
Carry Skip Adder



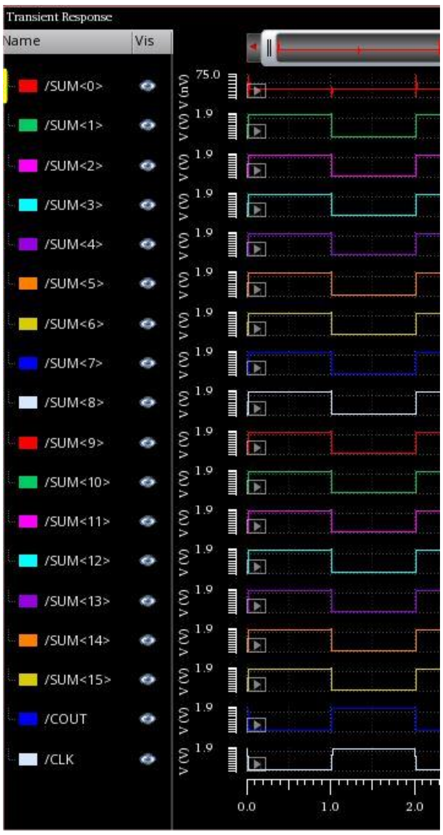
Binary value:

0000000001010101 + 0000000011001100 = **0100100001**

Manchester Carry Chain Adder



Carry Skip Adder

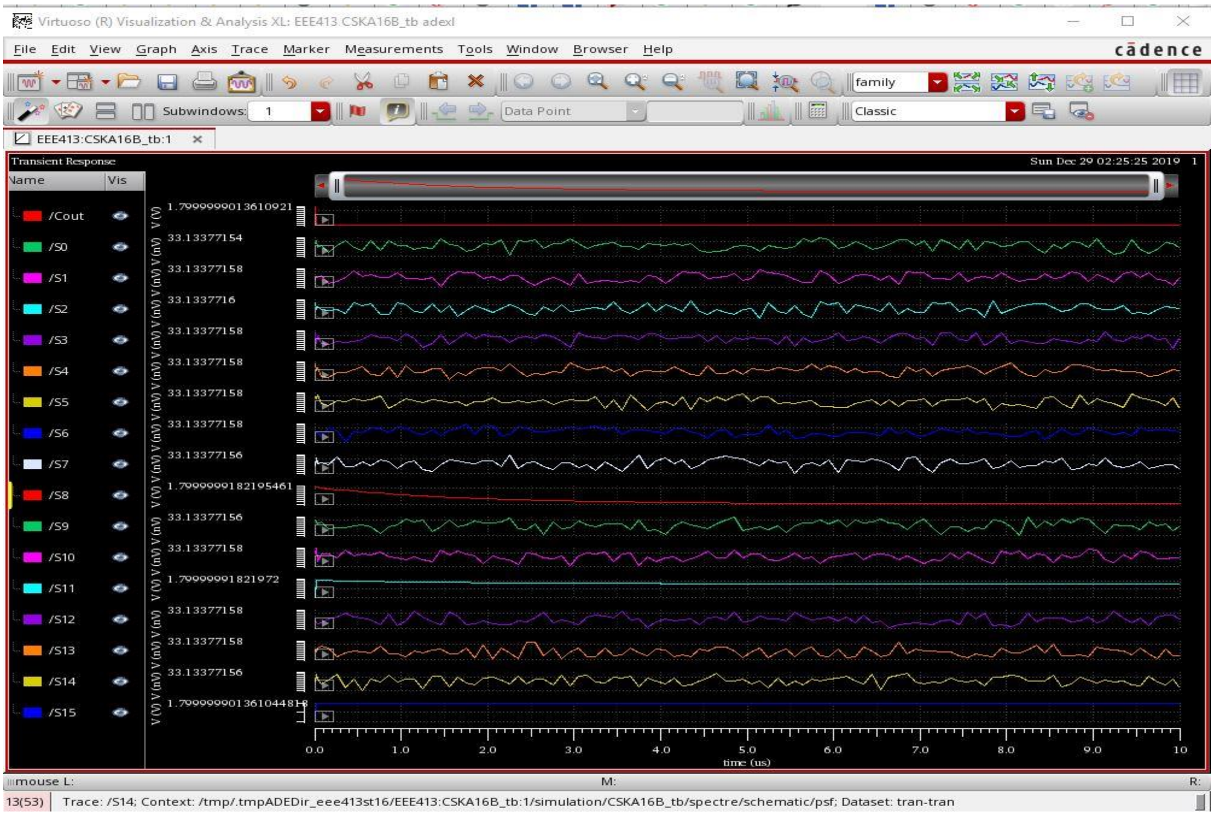


Binary value:

1111111111111111 + 0000000000000001

= 10000000000000000

Manchester Carry Chain Adder



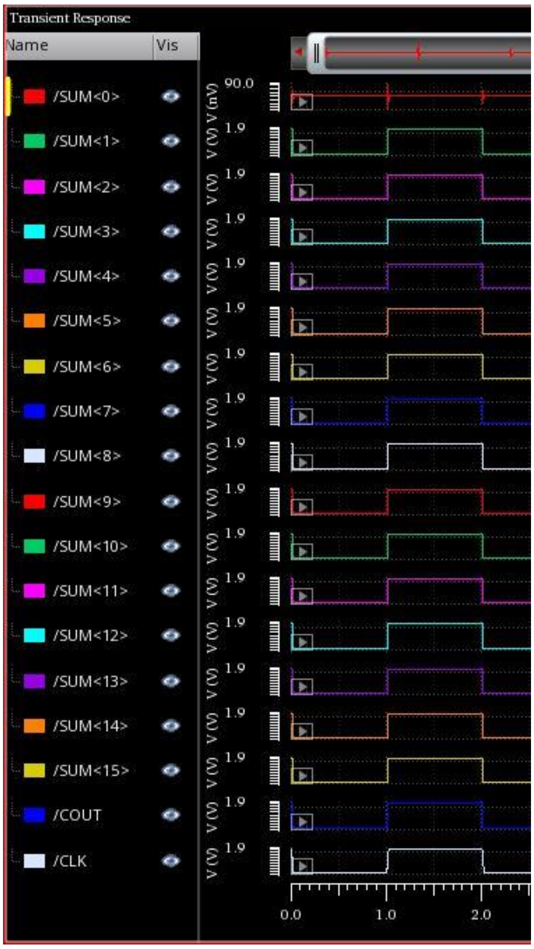
Carry Skip Adder

Binary value:

1001100100000000 + 1111000000000000 = 01100010010000000

Manchester Carry Chain Adder

Carry Skip Adder

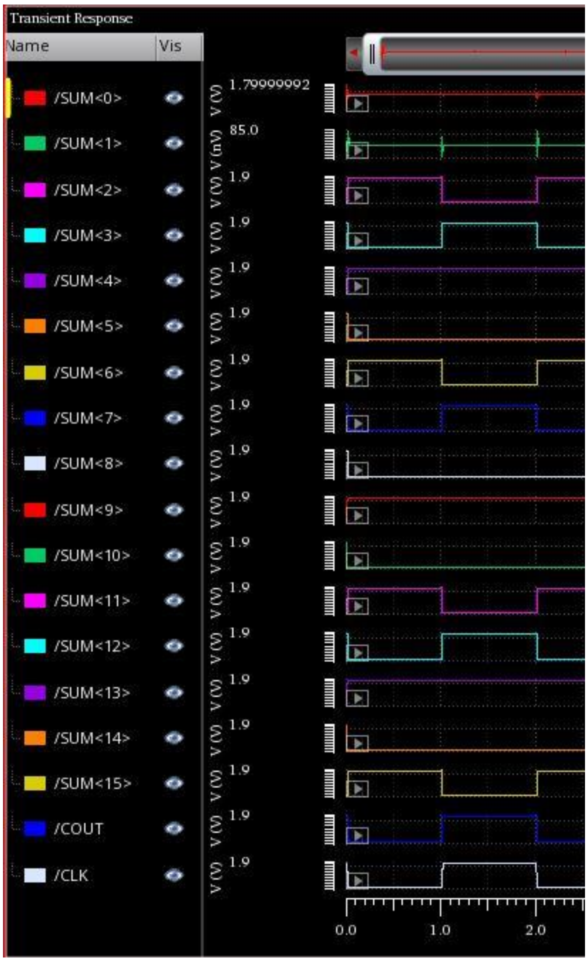


Binary value:

1111111111111111 + 1111111111111111 = 01111111111111110

Manchester Carry Chain Adder

Carry Skip Adder



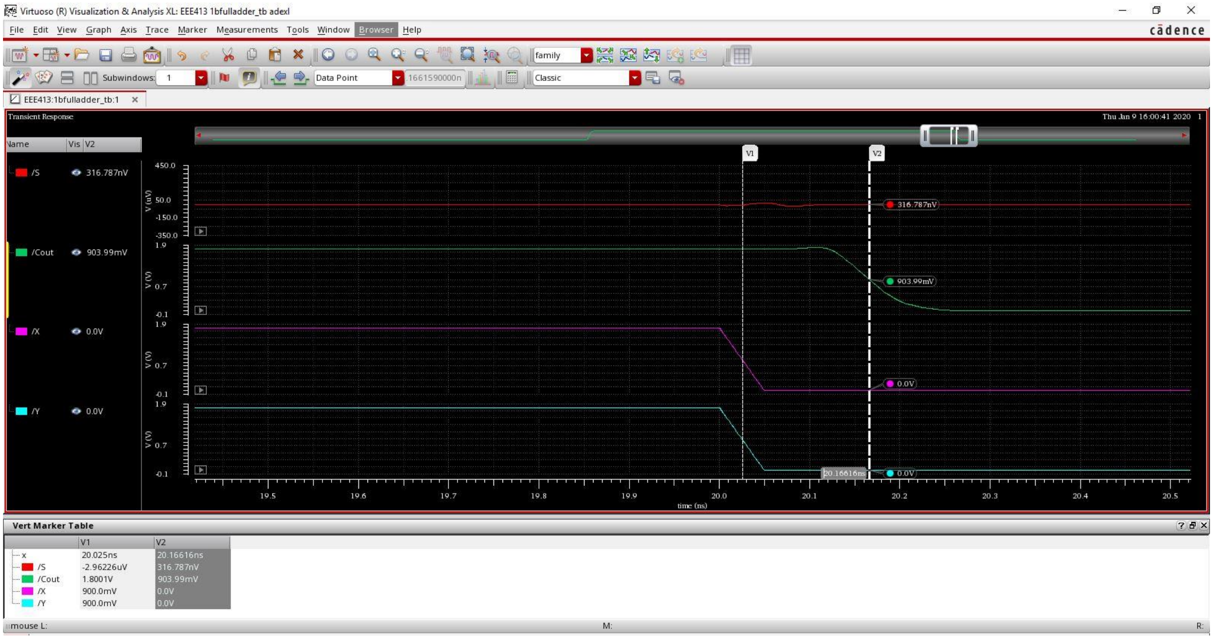
Binary value:

1100110000110011 + 0110011001100110
= **0100110010100111001**

Manchester Carry Chain Adder

Worst Case Delay:

CARRY SKIP ADDER:



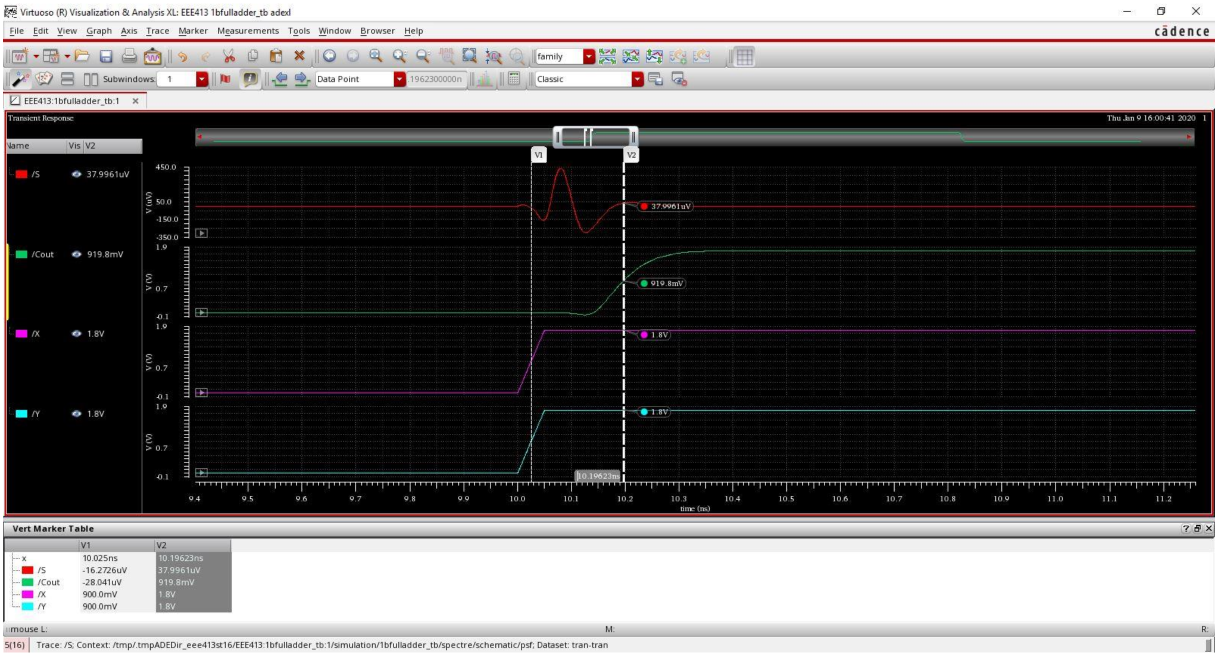


Figure 13- 1 bit full adder HL-LH delays

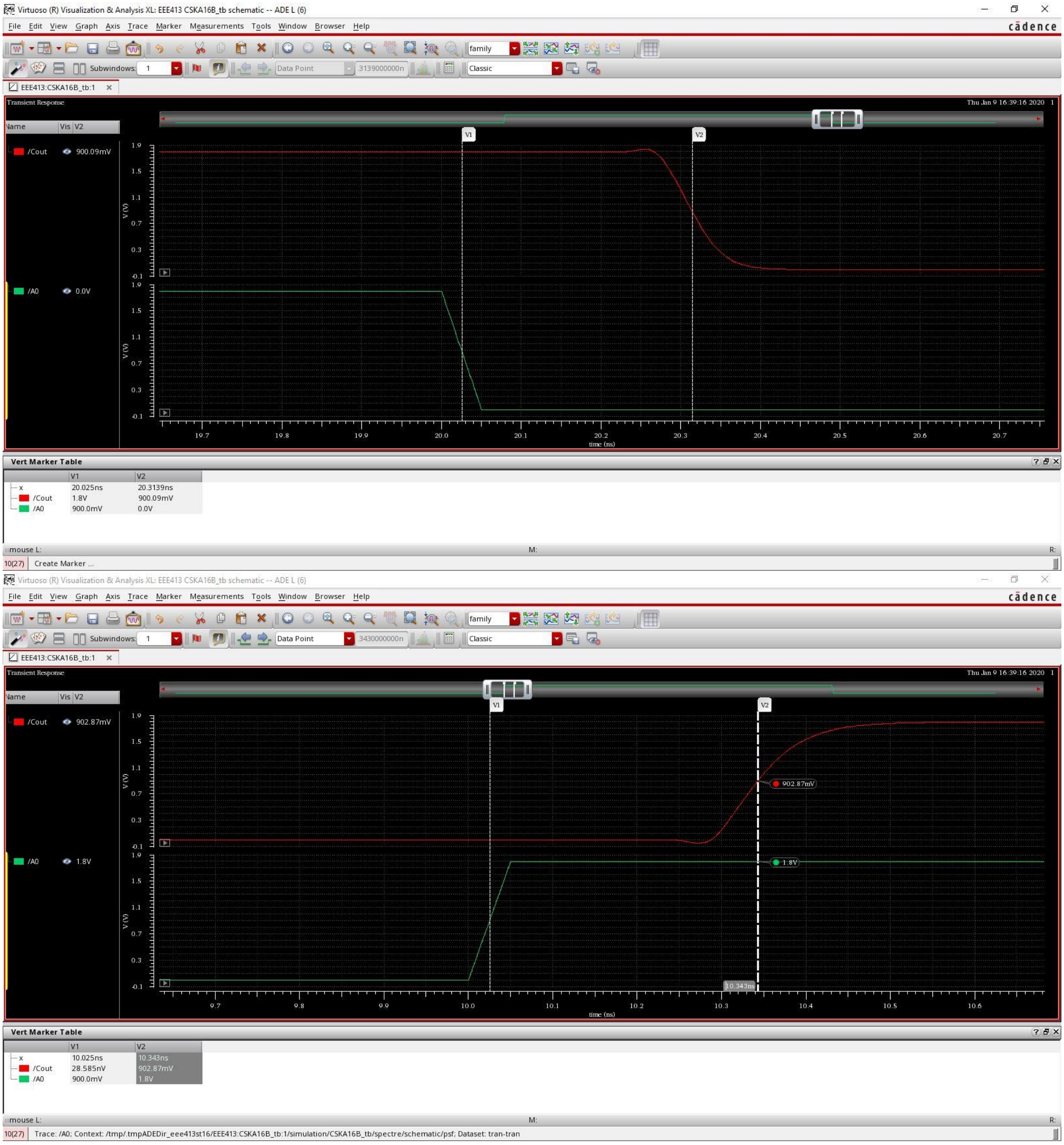


Figure 14 - 16 bit carry skip adder HL-LH delays

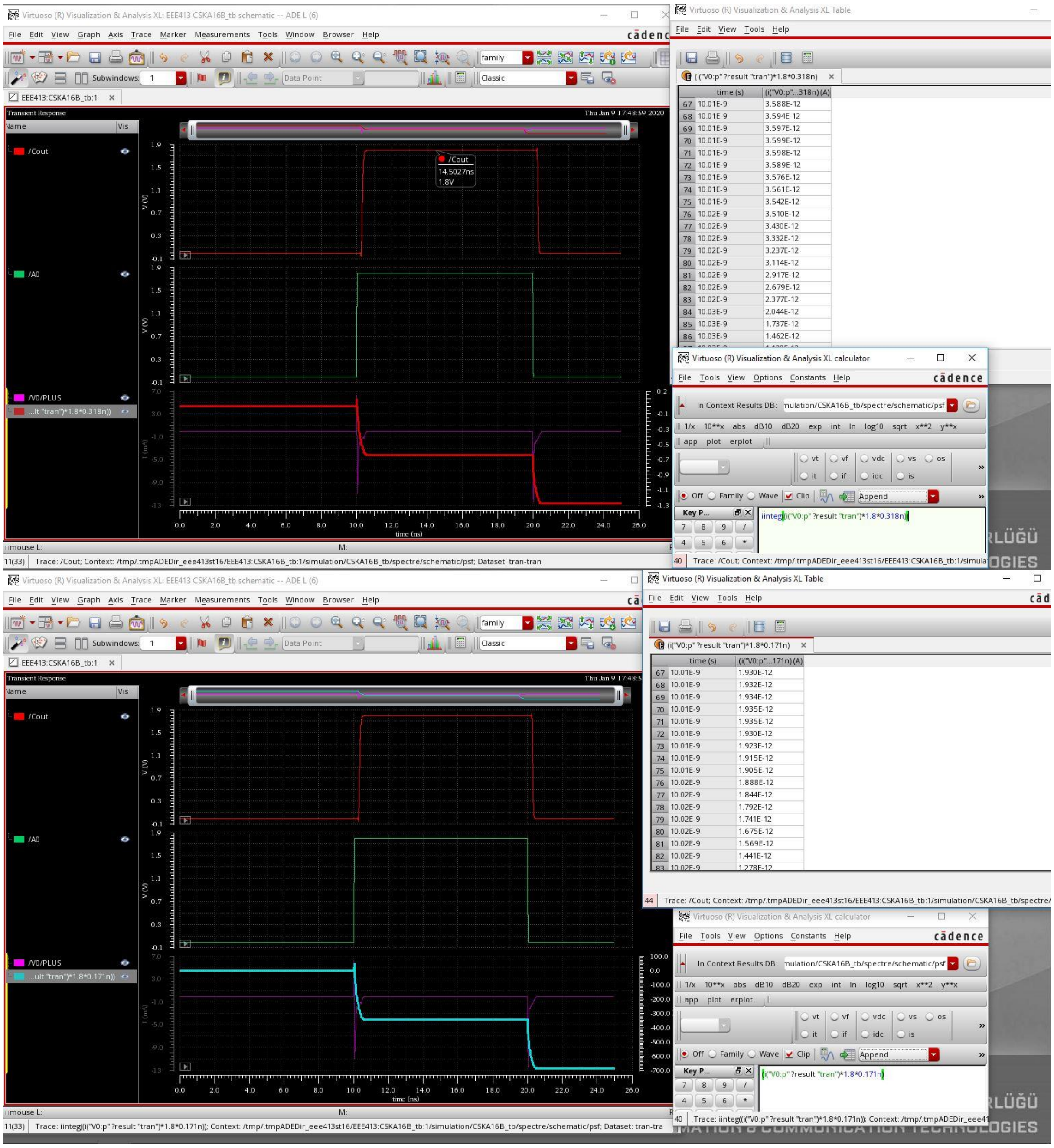


Figure 15 - 16 bit carry skip adder HL-LH energy delay product

MANCHESTER CARRY CHAIN ADDER:

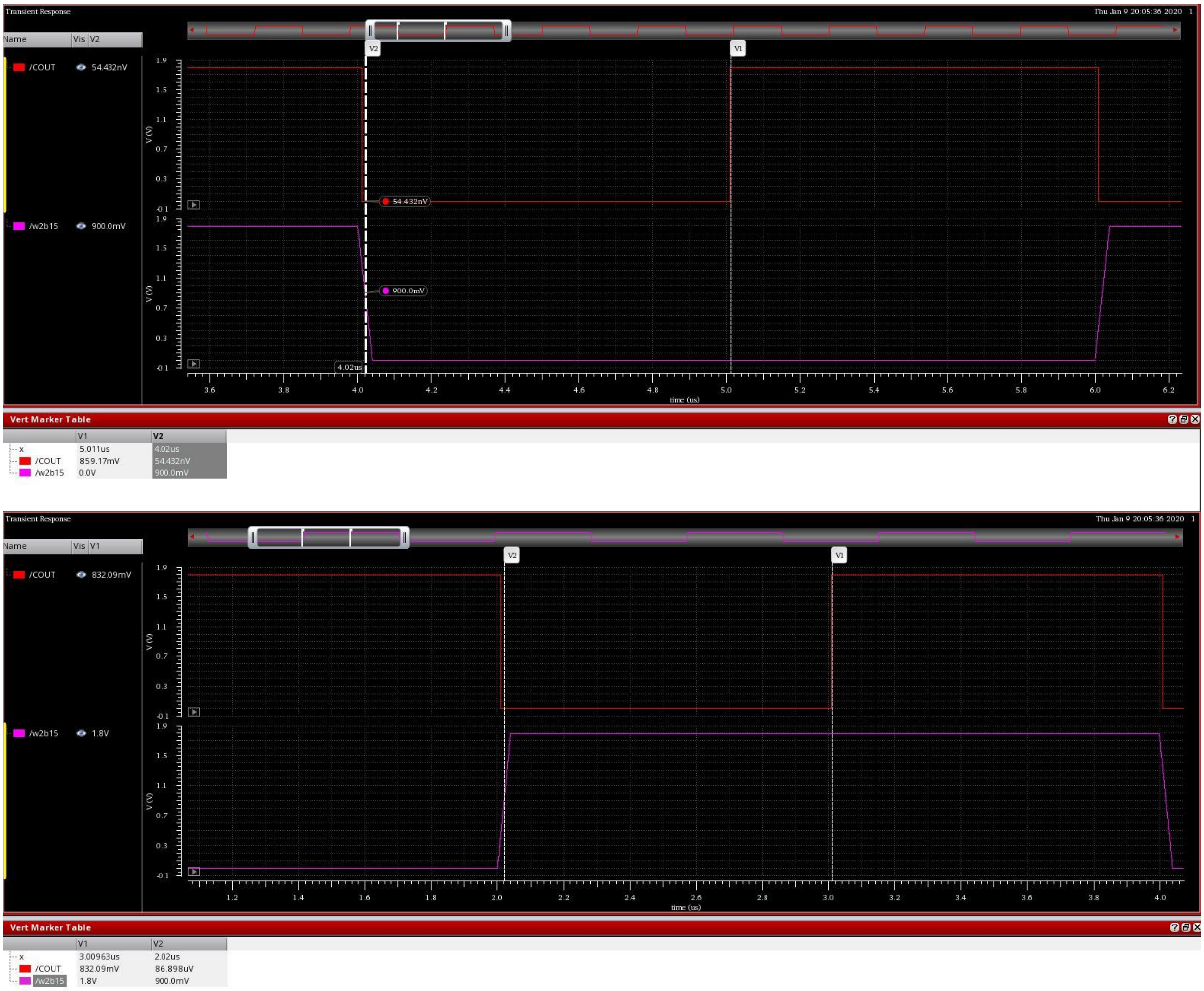


Figure 16 - MCCA HL-LH delays

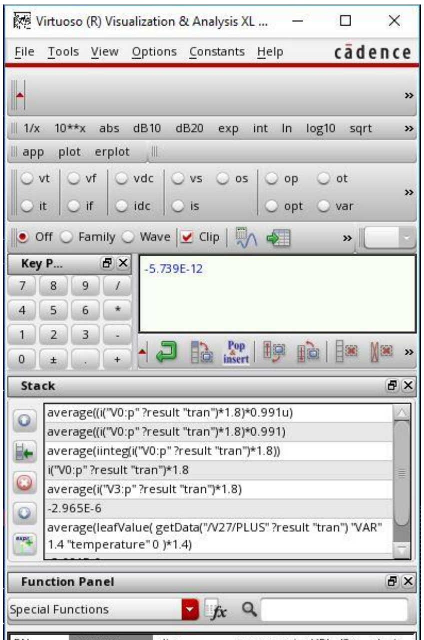


Figure 17-MCCA HL Energy concumption

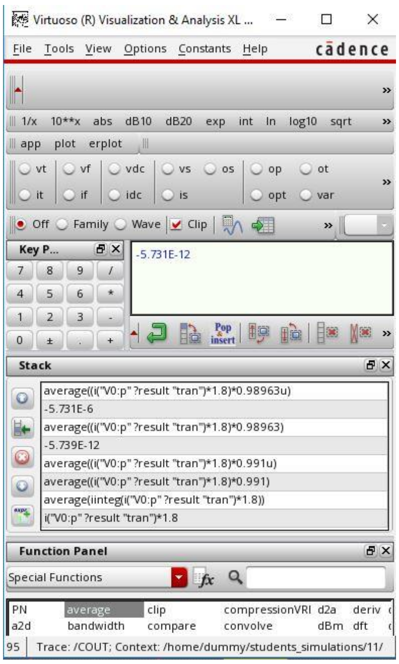


Figure 18 - LH Energy Consumption

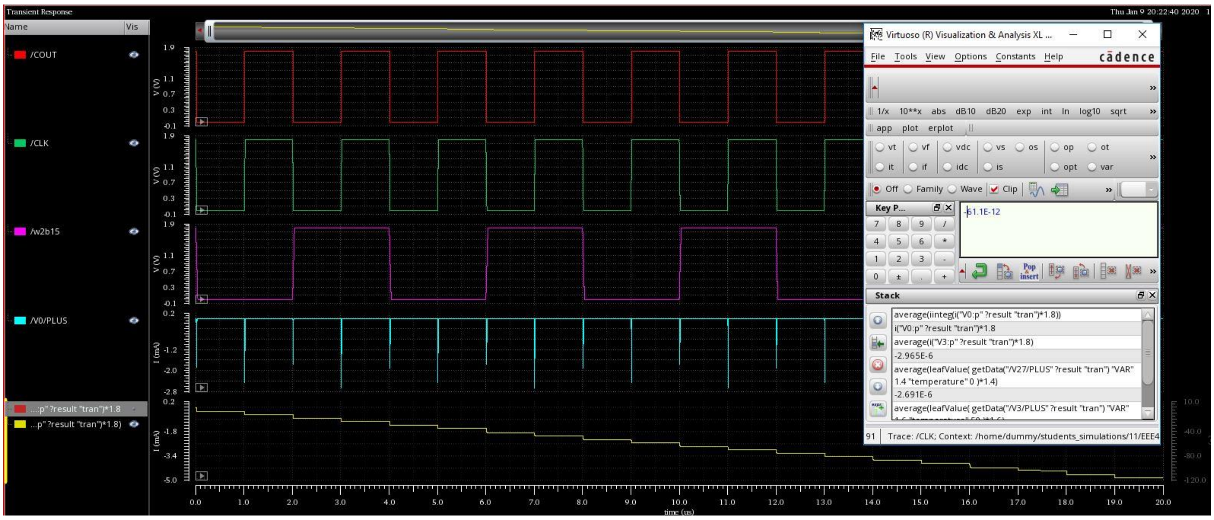


Figure 18 - MCCA Average Energy Consumption

For a CSKA that contains N cascaded full adders, the worst propagation delay of the summation of two N-bit numbers, A and B, belongs to the case where all the FAs are in the propagation mode. In other words, the worst case delay path occurs when the all inputs are 1.

Also in Manchester Carry Chain Adder, the worst case delay occurs when all of the inputs are HIGH aswell. LH and HL delays and energy consumption of the MCCA is calculated according to this fact.

Part V

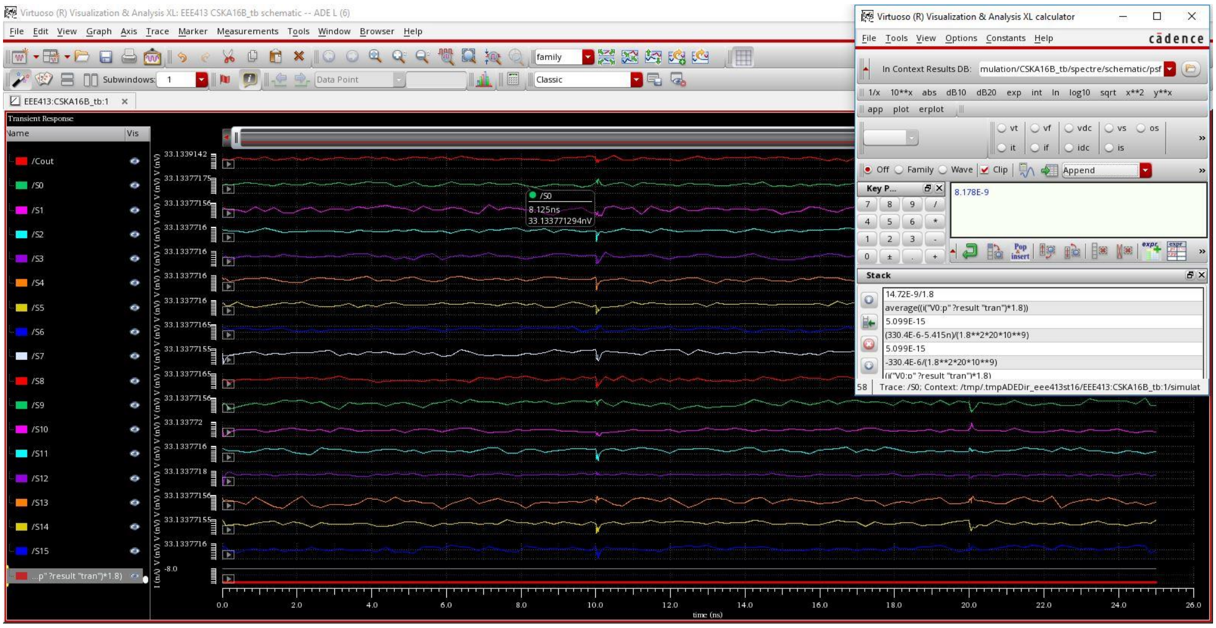


Figure 19 - Average Leakage Current for CSKA

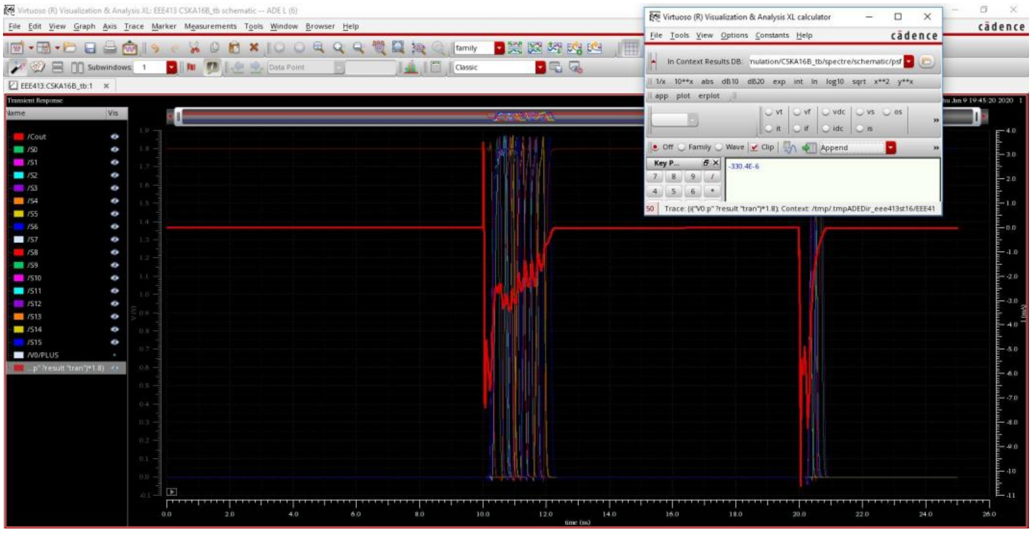


Figure 20 - Total power of CSKA

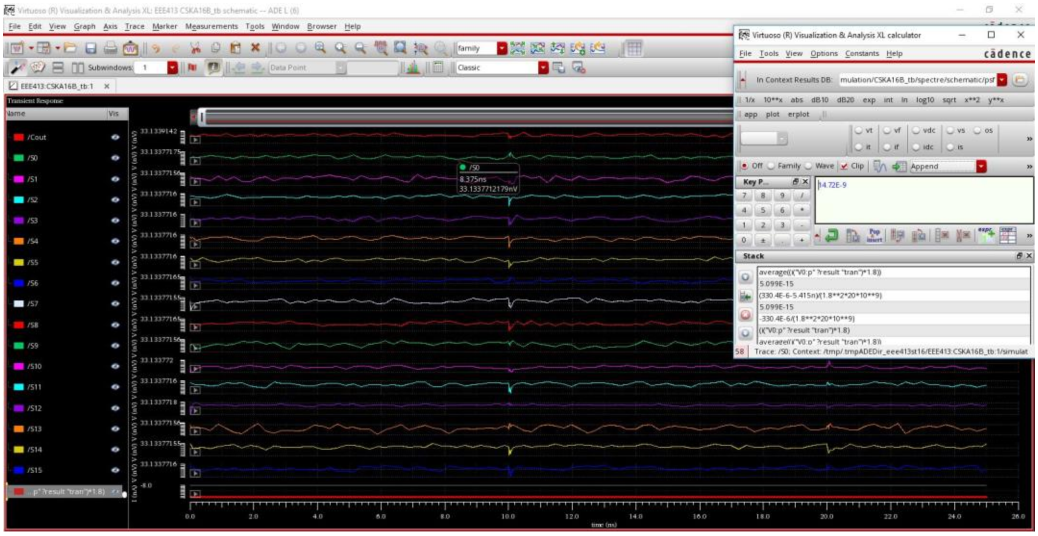


Figure 22- Average static power of CSKA

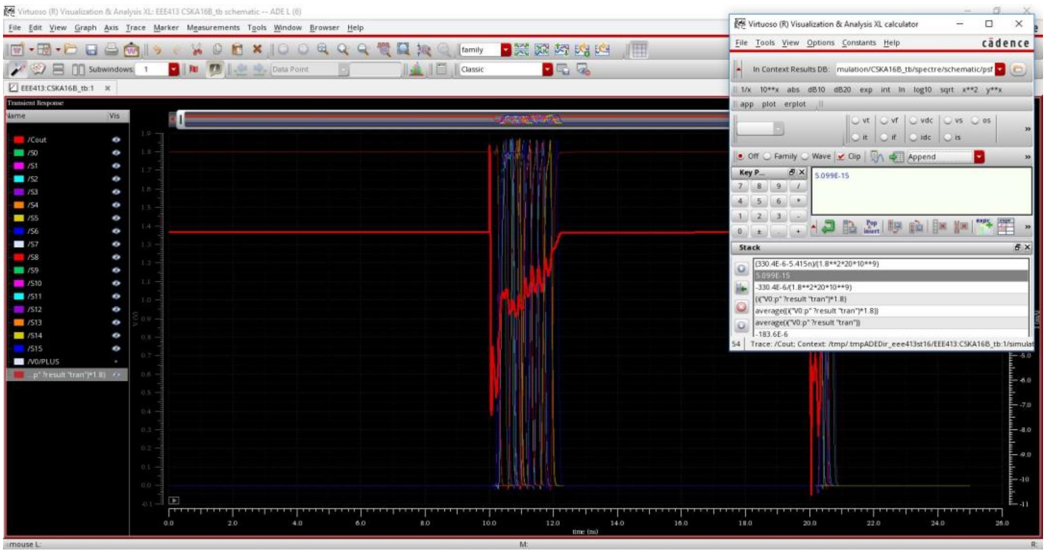


Figure 21- Dynamic Capacitance of CSKA

Figure 24- Average Leakage Current of MCCA

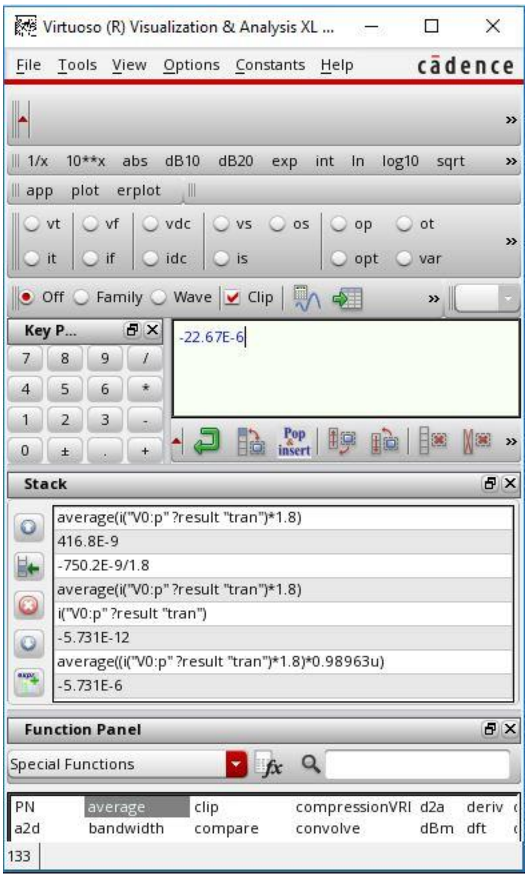


Figure 23 - Total power of MCCA

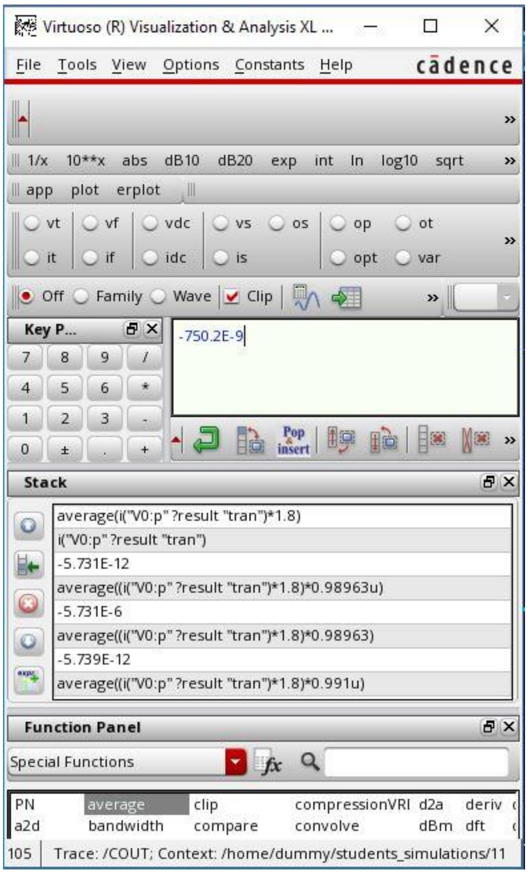


Figure 26 - Average static power of MCCA

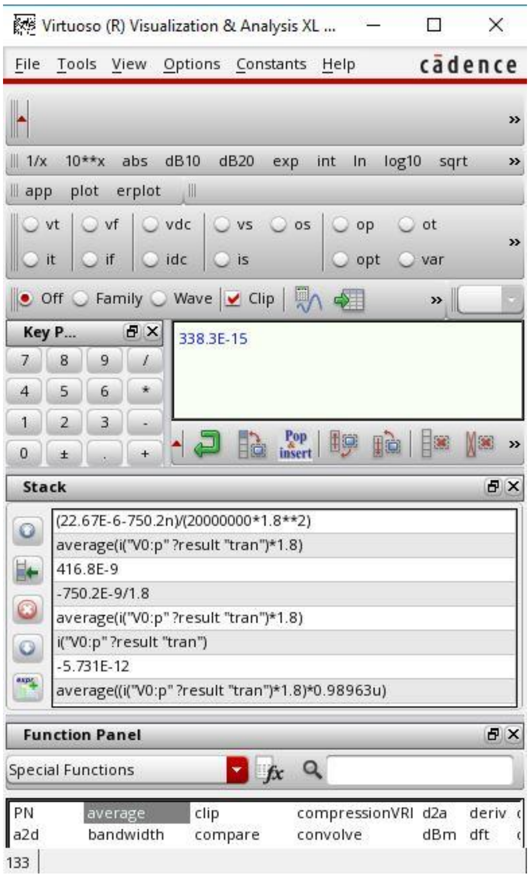


Figure 25 - Dynamic Capacitance of MCCA

The static power is found to be 14.72nano for the CSKA and 750 nano for the MCCA, after that Average Leakage current is calculated as 8.178nano for CSKA and 416.8 nano for MCCA. Later on total power is calculated and dynamic power is calculated by substituting static power from total power. 20GHz is used as switching frequency in both cases.

Dynamic power is divided to switching frequency and second power of Vdd then I found the average dynamic capacitance.

CONCLUSION

If we compare both adders, Carry Skip Adder is good for low power applications and Manchester Cary Skip Adder is good for high speed calculations. This fact is proven in our simulations aswell. Hardwarewise the advantage of the Manchester Carry Chain Adder is that it does not have any logic gates between the stages apart from the buffers connecting two 4-bit adders. In contrary to this, Carry Skip Adder has muxes, or gates and xor gates between the stages.

Although MCCA has lower delay and CSKA has lower power dissipation, calculations with higher number of bits results in low speed in MCCA and lower bit operations result in high delay and high power consumption in CSKA.

In addition Carry Skip Adder has larger layout area in comparison to the Manchester Carry Chain Adder.

In summary, Carry skip adder propagates the carry if and only if all of the inputs are HIGH for a two bit block. Additions occur using consecutive full adders. In Manchester carry chain, carry propagates accoding to the state of the PG block. Carry in and Carry out are inverted. Addition is done using an XOR block and because this particular adder is synchronous, it has precharge and evaluate states. Addition only occur in the evaluate state.